How much is *too much*?

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Semantics 4

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1 Review of strength-based maximality

In the generalized version of Sharvy's (1980) account, the definite article denotes the unique maximal element in in its set argument.

Von Fintel et al. (2014) offer an alternative to traditional maximality accounts by saying that definiteness targets the member of a set that yields the strongest proposition.

(1) the amount of nuts sufficient to make a pan of baklava (von Fintel et al. 2014: 168)

In strength-based maximality, the definite article refers to the minimally sufficient quantity, which yields the strongest/most informative proposition compatible with the context.

The predicate *sufficient to make a pan of baklava* is downward scalar in terms of semantic strength: the lower the amount that is sufficient, the stronger the proposition.

Similarly, as for the semantics of degree questions, Fox and Hackl (2007) modified Rullmann's (1995) account as follows.

(2) (Fox and Hackl 2007, item (22))
How much/many φ?
a. Rullmann (1995):
What is the maximal degree d s.t. φ(d)?
b. modification based on Beck and Rullmann (1999):
What is the degree d that yields the most informative among the true propositions of the form $\varphi(d)$?

(3) How many cups of nuts are sufficient to make a pan of baklava?

By using MAX_{inf}, both the definite description proposed by von Fintel et al. (2014) and the meaning of the degree question by Fox and Hackl (2007) lend support to a unified account between a maximality and a minimality effect.

2 Negative islands and the universal density of measurement (UDM)

What MAX_{inf} might not be able to account for while integrating a revised entry for questions and definite descriptions is the unacceptability of some constructions with negative islands.

(4) Questions

a. *How much water does this cup of koicha not have?

b. *How much does John not weigh?

c. *How many children does John not have?

(5) Definite descriptions

a. *the amount of water that this cup of koicha doesn't have

b. *the amount of nuts that I don't need

c. *the number of children that I don't have

Rullmann (1995) accounts for this unacceptability by arguing that degree questions (and definite descriptions) are blocked by negative islands, since negation interferes with the ability to identify a maximal degree required by these constructions.

UDM's approach to negative islands: maximality requirement, which maintains that for degree questions and expressions involving measurement, we need to identify a maximal degree that satisfies the truth conditions of the utterance.

(6) The Universal Density of Measurements (UDM) hypothesis (Fox and Hackl 2007)
Measurement scales needed for natural language semantics are *always* dense.

The UDM leads to certain predictions regarding sentences that involve negative islands. It also explains instances where *wh*-extraction across negation can occur as long as maximality does not create a contradiction.

(7) a. How much water are you <u>sure</u> that this cup of koicha does<u>n't</u> have?
b. How many children are you *not* <u>allowed</u> to have if you live in China?
c. the amount of radiation that you are *not* <u>allowed</u> to be exposed to
d. ?the amount of radiation that the company is <u>required</u> *not* to expose its workers to
e. *the amount of food that the company is [<u>allowed</u> *not*]/[*not* <u>required</u>] to give its workers
(the b-e examples are modified from Fox and Hackl 2007)

(8) (Suppose that there is 40 millilitres of water in the cup of koicha being said.
Given density, the complement set of the set of degrees *40* s.t. that the cup of koicha has at least *40* millilitres of water must be an *open* interval.)
**the amount of water that this cup of koicha doesn't have* → violation of max. requirement

(9) (It is possible for the set of degrees *d* s.t. there is *certainty* that this cup of koicha doesn't have at least *d* millilitres of water to be a *closed* interval)
the amount of water that you're sure this cup of koicha doesn't have → acceptable as there is a minimal degree member.

(10) Constraint on Interval Maximization (CIM): N-open monotone properties cannot be maximized by MAX_{inf}

The CIM helps account for various behaviours of degree constructions, demonstrating the relationship between the degree properties and the modal operators.

From F&H (2007):

(11) Basic consequence of the CIM
When φ denotes a property of degrees that is N-open and monotonic,
a. *$\text{wh}_\text{d}\ \varphi(d)$
b. *the $\lambda d.\varphi(d)$

(12) Consequence for universal modals
Since a universal modal has the capability to close an interval,
a. $\text{wh}_\text{d}\ \Box\varphi(d)$
b. the $\lambda d.\ \Box\varphi(d)$

(13) Consequence for existential modals
Since an existential modal cannot successfully close an interval,
a. *$\text{wh}_\text{d}\ \Diamond\varphi(d)$
b. *the $\lambda d.\Diamond\varphi(d)$

For the CIM to be applied, one must assume that there is a *formal* system that defines how a linguistic expression is evaluated without contextual dependency. F&H call it *deductive system* (DS).

The DS only considers dense scales as part of the formal semantic component, not discrete scales.

Non-formal considerations like cardinality is introduced by the parameter of granularity, which reflects the level of precision required in interpreting measurements or quantities.

(14) Granularity for the measurement of collections of indivisible objects (F&H 2007)
xGy iff there is a natural number n, s.t. $x \in (n, n+1]$ and $y \in (n, n+1]$.

3 *Too much*-predicate and the threshold of violation

Given the notion of UDM, I attempt to account for the (un)acceptability of the following instance.

(15) (??)the amount of water that is *too much* for a cup of koicha
(Suppose that the suggested amount of water for preparing a cup of koicha is 40 millilitres.)

In the DS, whether or not (15) is ruled out would depend not only on contradiction evaluation, but also how vaguely *too much* can be understood.

Without assessing how vague/precise the *too much*-predicate is, the DS would recognize that there are infinitely many amounts of water greater than any specified upper limit, and thus no minimal degree can be identified. (*too* may be perceived as a form of implicit negation?)

However, if an additional contextual constraint can resolve the vagueness (e.g., if *too much* is explicitly defined as any amount exceeding a <u>certain</u> limit), then the DS may accept (15)?

Can the DS incorporate pragmatic tolerance as a mechanism where speakers are not strictly bound by formal definitions in every context?

4 References

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